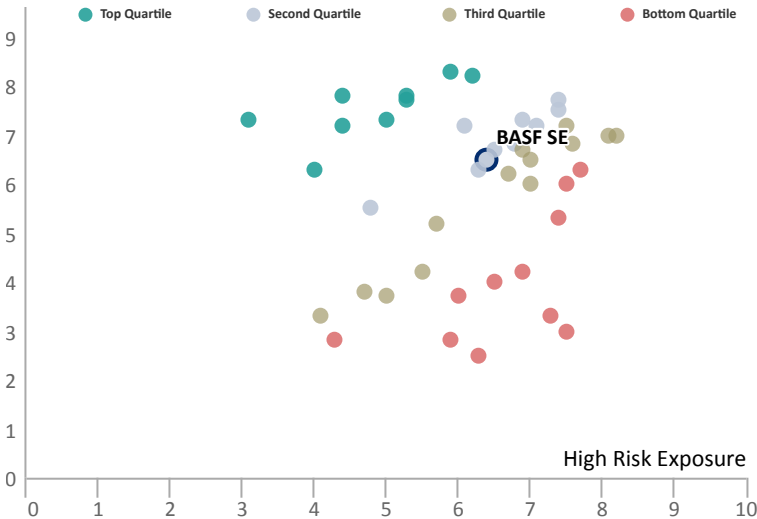




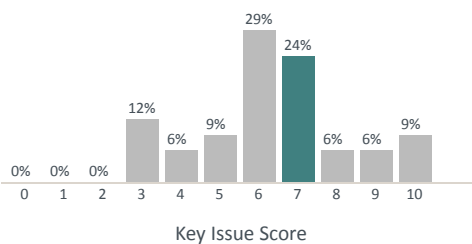
Carbon Emissions

Score	Change (since rating)	Quartile	Weight	Last score change date
7.1	0.0	●●●	13.0%	Sep 19, 2025

Strong Risk Management



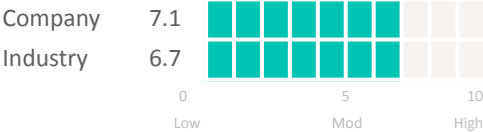
KEY ISSUE SCORE DISTRIBUTION*



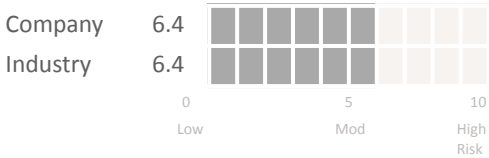
TOP 5 INDUSTRY LEADERS

LG CHEM LTD	10.0
THE SUPREME INDUSTRIES LIMITED	10.0
Saudi Basic Industries Corporation SJSC	9.8
ASAHI KASEI CORPORATION	9.3
NAN YA PLASTICS CORPORATION	9.3

KEY ISSUE ASSESSMENT



RISK EXPOSURE ASSESSMENT



Drivers of Risk Exposure

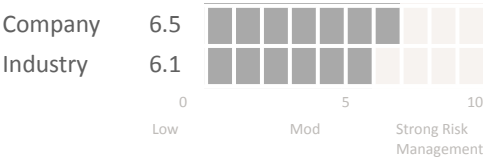
Business Types



Business Locations



RISK MANAGEMENT ASSESSMENT



Drivers of Risk Management

Practices Score



Performance Score



BOTTOM 5 INDUSTRY LAGGARDS

HENGLI PETROCHEMICAL CO., LTD.	4.3
Zhejiang Juhua Co., Ltd.	3.9
Jiangsu Eastern Shenghong Co., Ltd.	3.2
Huaфон Chemical Co., Ltd.	3.0
INNER MONGOLIA JUNZHENG ENERGY & CHEMICAL GROUP CO., LTD.	2.5

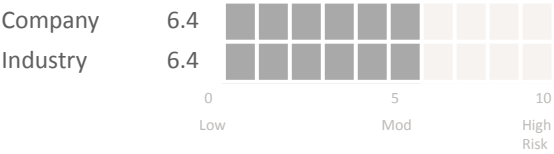
METHODOLOGY NOTE

Companies are evaluated on the carbon intensity of their operations and their efforts to manage climate-related risks and opportunities.

* [For symbols and terms used in this report, refer to the Glossary section at the end of the report]

EXPOSURE

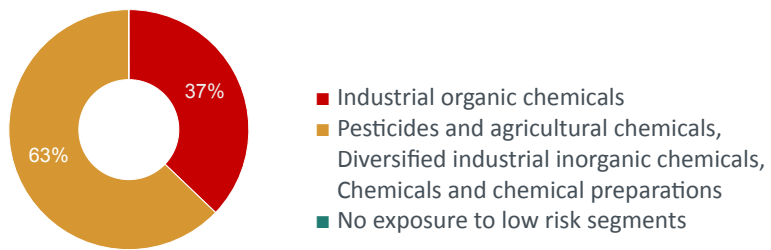
RISK EXPOSURE
ASSESSMENT



Key Drivers of Risk Exposure

Exposure to: Risks of having to pay increased compliance costs tied to carbon emissions regulations

Business Types : Percentage of operations in business segments with high/moderate/low carbon intensity



Source: IERS' Comprehensive Environmental Data Archive (CEDA); Air Emissions Accounts (Eurostat); Refinitiv; MSCI ESG Research; company disclosures

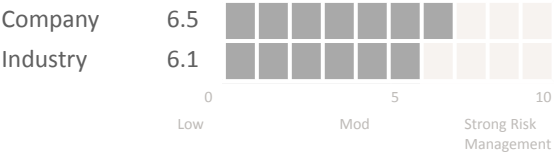
Business Locations : Percentage of operations in countries with strengthening or pending carbon emissions regulation



Source: MSCI ESG Research; Refinitiv; company disclosures

MANAGEMENT

RISK MANAGEMENT
ASSESSMENT



Description	Company Practice	Best Practice	Practices Score ²			
Targets						
Aggressiveness of the company's reduction target in the context of its current performance	Aggressive target with a low base	Aggressive target with a low base	-	LOW	MID	TOP
Demonstrated track record of achieving its carbon reduction targets	Not on pace to achieve current reduction target	Previously set & met targets	-	LOW	MID	TOP

Carbon Improvement Targets

Baseline Year	Target Year	Target Description	Target Reduction (%)
	2050	committed to achieve net zero by 2050 including scope 1,2,3 (category 1)	100.00%
	2050	We have set ourselves ambitious goals and are striving worldwide to achieve net zero scope 1 and 2 CO2 emissions by 2050	100.00%
	2025	Increase renewable electricity in zhanjiang verbund site by 100% by 2025	
	2030	We aim to source more than 60% of our power needs from renewable sources by 2030.	
	2030	Reduce CO2 emissions per ton of crop by 30% by 2030	30.00%
	2050	Achieve climate neutral by 2050	100.00%
2022	2030	reduce scope 3 intensity emissions (metric tons of co2 per ton of raw material bought) by 15% by 2030 from 2022 base year including purchased goods and services	15.00%
2018	2030	We aim to maintain total greenhouse gas emissions from our production sites and our energy purchases at the 2018 level by 2030	0.00%
2018	2030	Carbon Neutrality (GHG emissions reduction from production site excluding sale of energy to third parties and purchase of own energy	100.00%
2018	2030	Reduction of 25% in our absolute greenhouse gas emissions by 2030 compared with 2018	25.00%
2018	2030	To reduce carbon emission absolute on scope 1 and 2 market based by 25% by 2030 based on 2018 baseline year	25.00%
2018	2030	reduction of scope 1,2, market based by 25% by 2030 from 2018 base year	25.00%
2002	2020	Scope 1&2 (location-based)reduction on GHG emissions per ton of sales product (excl. Oil & Gas)	40.00%
2002	2020	Increase its energy efficiency of production processes by 35% previously 25%	35.00%

Mitigation

Strength of Greenhouse Gas Mitigation Strategy (0-10 Score, 0=worst, 10=best) 7.00

Programs or actions to reduce the emissions intensity of core operations

Use of cleaner sources of energy	Some efforts	Aggressive efforts	-	LOW	MID	TOP
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Capture GHG emissions	Some efforts	Aggressive efforts	-	LOW	MID	TOP
Energy consumption management and operational efficiency enhancements	Some efforts	Aggressive efforts	-	LOW	MID	TOP
CDP disclosure	Yes	Yes	-	LOW		TOP

Performance

Carbon Emissions Performance Relative to Peers (0-10 Score, 0=worst, 10=best) 6.00

GHG Emissions - metric tons CO2e

Year	Scope 1 Disclosed	Scope 1 Estimate Key	Scope 2 Disclosed	Scope 2 Estimate Key	Scope 1+2 Disclosed	Scope 3 (upstream)	Scope 3 (downstream)	Scope 3 (undefined)	Scope 1 Estimated	Scope 2 Estimated	Scope 1+2 Estimated	Scope 1+2 Estimate Key	GHG Emissions Details
2024	15,216,000.0	Reported	3,631,000.0	Reported	18,847,000.0	62,100,000.0	32,410,000.0					Reported	
2023	15,562,000.0	Reported	3,317,000.0	Reported	18,879,000.0	54,363,000.0	31,072,000.0					Reported	
2022	16,556,000.0	Reported	3,588,000.0	Reported	20,144,000.0	58,104,000.0	34,092,000.0					Reported	
2021	18,667,000.0	Reported	3,670,000.0	Reported	22,337,000.0	64,131,000.0	37,265,000.0					Reported	
2020	18,368,000.0	Reported	3,362,000.0	Reported	21,730,000.0	56,749,000.0	34,637,000.0					Reported	
2019	17,337,000.0	Reported	3,552,000.0	Reported	20,889,000.0	59,056,000.0	40,620,000.0					Reported	
2018	18,418,000.0	Reported	3,361,000.0	Reported	21,779,000.0	56,727,000.0	61,238,000.0					Reported	
2017	18,775,000.0	Reported	3,796,000.0	Reported	22,571,000.0	63,486,000.0	67,037,000.0					Reported	
2016	18,036,000.0	Reported	3,884,000.0	Reported	21,920,000.0	65,031,000.0	68,172,000.0					Reported	
2015	18,375,000.0	Reported	3,795,000.0	Reported	22,170,000.0	58,635,000.0	64,195,000.0					Reported	
2014	18,450,000.0	Reported	3,911,000.0	Reported	22,361,000.0	60,444,000.0	76,358,000.0					Reported	
2013	18,842,000.0	Reported	3,987,000.0	Reported	22,829,000.0	58,630,000.0	67,067,000.0					Reported	
2012	20,208,000.0	Reported	4,479,000.0	Reported	24,687,000.0	54,010,000.0	75,152,000.0					Reported	
2011	20,920,000.0	Reported	4,879,000.0	Reported	25,799,000.0	71,180,000.0	80,760,000.0					Reported	
2010	21,312,000.0	Reported	4,389,000.0	Reported	25,701,000.0							Reported	
2009	27,523,000.0	Reported	4,108,000.0	Reported	31,631,000.0							Reported	
2008		E.CSI		E.CSI					19,924,326.359733738	4,110,539.6265753694	24,034,865.986309107	E.CSI	May 2025 Emissions Estimation Model Update

GHG Emissions Intensity - metric tons CO2e / USD million sales

Year	GHG Intensity	GHG Intensity Details	GHG Intensity - Reported	GHG Intensity - Reported Details
2024	278.90			
2023	248.04			
2022	216.14			
2021	249.91			
2020	300.26			
2019	313.73			
2018	303.98		188.40	metric tonnes CO2e / full time equivalent (FTE) employee
2017	291.53		199.20	metric tonnes CO2e / full time equivalent (FTE) employee
2016	361.08		196.30	metric tonnes CO2e / full time equivalent (FTE) employee
2015	289.69			

Year	GHG Intensity	GHG Intensity Details	GHG Intensity - Reported	GHG Intensity - Reported Details
2014	248.63			
2013	224.67			
2012	237.69			
2011	270.99			
2010	300.57			
2009	435.53			
2008	277.51			

Energy Consumption

Year	Total energy consumption (reported)	Total energy consumption (MWh)	Energy intensity (reported)	Reported intensity details	Total energy consumption intensity	Energy consumption details
2024	75,600,000	75,600,000	1.16	MWh Million/ billion €	1,118.73	Total energy consumption
2023	119,500,000	119,500,000			1,570.04	total energy consumption
2022	52,900,000	52,900,000			567.60	Total Energy consumption
2021	60,840,000	60,840,000			680.68	Electricity + Fuels + Steam
2020	61,253,500	61,253,500			846.37	Electricity + Fuels + Steam
2019	59,166,080	59,166,080			888.62	Electricity + Fuels + Steam
2018	60,618,360	60,618,360			846.07	Electricity + Fuels + Steam
2017	61,280,000	61,280,000			791.51	Electricity + Fuels + Steam
2016	86,346,000	86,346,000			1,422.35	Electricity + Fuels + Steam
2015	85,051,000	85,051,000			1,111.36	Electricity + Fuels + Steam
2014	77,204,000	77,204,000			858.41	Electricity + Fuels + Steam
2013	76,609,000	76,609,000			753.94	Electricity + Fuels + Steam
2012	83,670,000	83,670,000			805.57	Electricity + Fuels + Steam
2011	79,128,000	79,128,000			831.15	Electricity + Fuels + Steam

Year	Total energy consumption (reported)	Total energy consumption (MWh)	Energy intensity (reported)	Reported intensity details	Total energy consumption intensity	Energy consumption details
2010	65,193,000	65,193,000			762.44	Electricity + Fuels + Steam
2009	61,409,000	61,409,000			845.55	Electricity + Fuels + Steam